Team Project Final Report - Montgomery County Crime Database

INST 327-0202

Montgomery County Crime Database Analysis

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May, 15, 2025

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Introduction

In today's world, data plays a critical role in understanding public safety and shaping just policy. For our final project, we chose to build a crime database focused on Montgomery County, Maryland. We aimed to create a transparent and analytical system for examining local crime patterns that could be ethically used by communities, researchers, and policymakers. Crime data can too easily be misrepresented or misused when not structured properly, especially when demographic attributes like race or socioeconomic status are involved. With this in mind, we focused on designing a schema that supports equitable and privacy-conscious analysis, avoiding personally identifiable information and potentially biased attributes while still allowing for insight into trends such as offense types, affected areas, and policing zones.

This database serves as an example of how thoughtfully organized data can become a tool for accountability and awareness. It empowers users to ask meaningful questions such as where crimes are most common, what types occur most often, and how incidents vary by location without reinforcing systemic biases. Our work covers the full lifecycle: from logical design and physical implementation to data cleaning, population, and query construction.

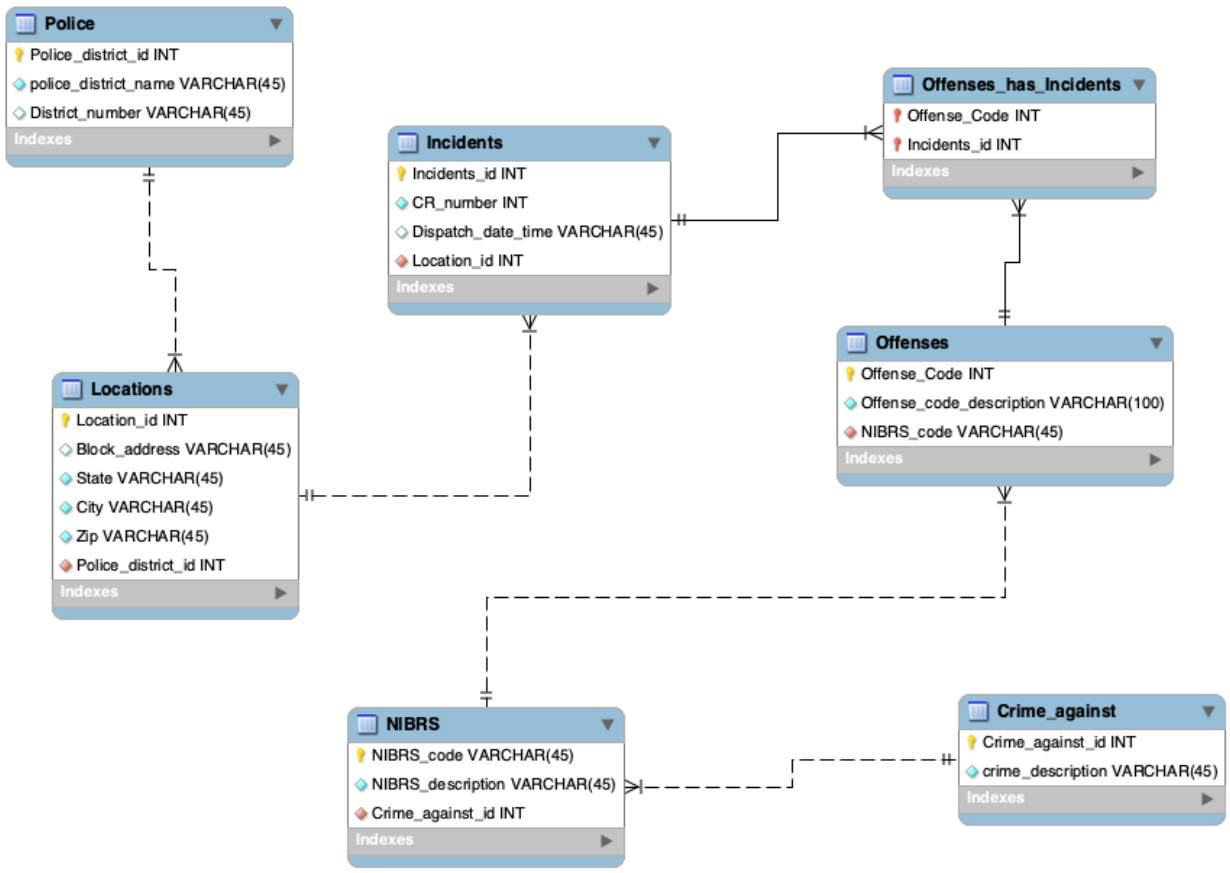
Database Description

The Montgomery County Crime database was structured around seven interrelated tables designed to represent the core elements of a crime incident report. These include information about the incident itself, the nature of the offense, the geographic details of the crime, and classification identifiers derived from standardized reporting systems. The Incidents table records individual crime reports, including identifiers, dates, and links to geographic data. The Offenses table captures the type and description of each reported offense. The NIBRS table provides a standard classification framework and links to broader categories found in the Crime_Against table, which labels crimes by their general type, against people, property, or society. Locations include block addresses, city, ZIP code, and district assignment. The Police table stores information about the policing jurisdiction responsible for the crime location. Lastly, the Incident_Offense_Link table serves as a junction table, associating incidents with one or more reported offenses.

The data used in this database was obtained from Montgomery County's public safety open data portal. Following initial filtering and deduplication, the dataset was trimmed to include fifteen representative records per base table to align with updated project scope requirements. The filtered records were then formatted and cleaned for normalization.

Logical Design

The final ERD, presented below, illustrates the logical design of our database:



This diagram demonstrates the one-to-many relationships between entities, clearly defines primary and foreign keys, and outlines the modular and normalized structure of our schema. Our design ensures referential integrity and was created with maintainability and scalability in mind. We prioritized clarity in table relationships and consistency in naming conventions, particularly when establishing foreign key connections. Special attention was given

to removing redundancy while ensuring each table had meaningful, non-null attributes that would support realistic queries.

Physical Database

We implemented the database physically using MySQL Workbench. Initially, we encountered a number of issues including improperly ordered table creation, incompatible data types (e.g., foreign keys referencing TEXT columns), and foreign key constraint failures. Based on TA feedback, we restructured our schema to address these problems. The corrected version included consistent primary and foreign key data types and ensured that every referenced table was created prior to the table using it. After reorganization, we executed the corrected schema successfully and populated the database using our filtered dataset. The physical database file, including tables, sample data, and saved views, was exported into a complete .sql backup file for submission.

Sample Data

The sample data used for this project was taken from Montgomery County's publicly available crime dataset. We used Excel to clean and organize the data. This included removing duplicate values, filtering out unsupported columns, and selecting representative records. The filtered dataset includes at least fifteen rows per base table and over thirty entries in the junction table. To ensure the data reflected realistic patterns, we applied several criteria. We prioritized crime reports from a mix of geographic regions including urban and suburban ZIP codes (e.g., Rockville, Wheaton, and Silver Spring), and we selected offenses that are common and clearly categorized under NIBRS standards (e.g., Auto Theft, Aggravated Assault, Credit Card Fraud). Our selection also focused on including variation in crime types (property, person, society) and associated policing districts. This balanced approach helps support analysis queries without overwhelming the schema.

Below is a sample of the structured data:

Incident ID	Offense Code	Crime Name3	Victims	Zip	Police District
201094549	2404	Auto Theft	1	20861	Wheaton
201233480	1302	Assault - Aggravated - Family - other Weapon	1	20850	Rockville
201362256	2605	Fraud - illegal use credit cards	1	20814	Bethesda
201304123	1207	Robbery - Residential - Gun	1	20866	Silver Spring
201397608	1399	Assault - 2nd degree	3	20877	Montgomery Village

Views and Queries

We developed five SQL queries that were saved as views to provide sample analyses on the data. These views demonstrate a range of SQL capabilities such as joins, filtering, aggregation, subqueries, and the use of linking tables.

The table below summarizes the queries and how they align with the required criteria:

View Name	Req. A (JOIN)	Req. B (FILTER)	Req. C (AGG)	Req. D (LINK)	Req. E (Subquery)
crime_by_zip	X	X		X	
top_crime_types	X	X	X		

victome_count_summary	X		X		
frequent_crime_areas	X	X	X	X	
crime_vs_district_avg	X		X		X

The following list describes what each query we wrote for our database displays:

Query 1: This query joins the Police, Locations, and Incidents tables to count the number of incidents that occurred within each police district. It groups the results by police district name and orders them by the total number of incidents in descending order. The view IncidentsPerDistrict saves this result for easy future access.

Query 2: This query joins NIBRS, Offenses, and Offenses_has_Incidents to find the frequency of each NIBRS crime description. It groups by the description, orders by frequency in descending order, and limits the result to the top 3. The view Top3FrequentCrimes stores this information.

Query 3: This query joins Locations and Incidents to count the number of incidents that occurred in each city. It groups the results by city and orders them by the number of incidents. The view IncidentsPerCity provides a quick overview of crime distribution by city.

Query 4: This query determines the most frequently occurring crime_description from the Crime_against table by joining it with NIBRS, Offenses, and Offenses_has_Incidents to count the number of associated incidents. It groups by the crime description, orders by frequency, and limits the result to the top one. The view MostCommonCrimeAgainst stores this single most frequent crime category.

Query 5: This query returns the number of distinct offense types committed within each police district. It uses multiple joins across Police, Locations, Incidents, Offenses_has_Incidents, and Offenses to count unique offenses per district. This view helps identify which police districts encounter the greatest variety of crime types, offering insight into the complexity and diversity of incidents handled by each district.

Changes from Original Design

Several changes were made to our design after receiving initial feedback. We increased the number from fifteen to thirty tables to meet the minimum criteria for our main tables and the linking tables. We removed unsupported components such as a proposed Case_Status table and dropped wireframe references. Additionally, we cleaned and standardized foreign key column names and restructured the order of our CREATE TABLE statements to avoid key-related execution errors. The ERD was also updated to reflect the corrected relationships. One specific correction was ensuring that crime_against_id was a foreign key in the NIBRS table (instead of the reverse) and that police district IDs were directly related to locations. Lastly, We had to change the data type of the NIBRS table, as the primary key was only accepting an integer data type. It became an issue when we tried to import the table because this primary key was composed of letters and numbers. The best way to tackle this problem was to make the primary key a VARCHAR data type.

Database Ethics Considerations

In designing our database, we ensured that ethical principles were respected at every stage. No personal identifiable information (PII) was included in our dataset, and all individual references were anonymized or abstracted. We intentionally omitted race, gender, and similar fields that could introduce bias into queries or reporting. By focusing solely on offense types, locations, and police jurisdictions, our database avoids enabling discriminatory insights and instead supports equitable policy analysis. All table relationships are transparent and clearly labeled to prevent misuse or confusion.

Lessons Learned

This project highlighted the importance of thoughtful design, collaboration, and iteration. We learned that foreign key relationships are not only conceptually important but must be defined precisely in the correct order to avoid errors during database creation. We also experienced how even a small dataset can uncover structural flaws, which forced us to iterate on our schema design and improve consistency. MySQL Workbench presented unexpected technical

hurdles, such as issues with using TEXT types in keys, but these were resolved by adjusting to VARCHAR types and testing repeatedly. Communication within our team was vital, especially when troubleshooting table relationships and refining our ERD.

Potential Future Work

If given additional time and resources, we would expand the database to include a larger dataset spanning multiple years of crime records. Automating the data cleaning and import process with Python scripts would also streamline future updates. Additional views and dashboards could be developed to provide real-time insights and geospatial mapping. We also envision connecting this database to external datasets, such as income levels or education statistics, to enable more comprehensive policy analysis. A front-end user interface could also be added for public access or police department internal tools.